LESSON 6.9

Divide Whole Numbers by Fractions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing a whole number by a fraction means — how many pieces fit, or how much is in one whole — find the quotient with a model or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Park has 4 pounds of evidence powder. Each testing bag holds exactly 1/2 pound. He needs to prepare as many testing bags as possible before the lab deadline. How many bags can he fill?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Whole Number *Spanish: Número entero*	A counting number with no fraction or decimal, like 0, 1, 2, 3.	4 is a whole number; as a fraction it is 4/1
Fraction *Spanish: Fracción*	A number that shows part of a whole, like 3/4.	1/2 means 1 out of 2 equal parts — like cutting a sandwich in half and taking one piece
Reciprocal *Spanish: Recíproco*	A fraction turned upside down.	The reciprocal of 2/3 is 3/2. Flip the fraction to divide: 6 ÷ 2/3 = 6 × 3/2
Keep, Change, Flip *Spanish: Mantener, cambiar, invertir*	A shortcut for dividing fractions: keep the first number, change ÷ to ×, flip the second. It works because dividing by a fraction gives the same answer as multiplying by its reciprocal.	6 ÷ 1/3 → Keep 6, Change ÷ to ×, Flip 1/3 to 3/1 → 6 × 3 = 18
Quotient *Spanish: Cociente*	The answer when you divide.	In 4 ÷ 1/2 = 8, the quotient is 8

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Agent Park has 4 pounds of powder and each bag holds 1/2 pound. I want 4 ÷ 1/2.

1. Each pound fills 2 half-pound bags, so 4 pounds fill 4 × 2 = 8 bags.
2. That is the same as multiplying by the reciprocal of 1/2: 4 × 2/1 = 8. Dividing by 1/2 doubles the count, because every whole holds 2 halves.
3. Check by multiplying: 8 bags × 1/2 pound = 4 pounds. So 4 ÷ 1/2 = 8. He can fill 8 bags.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 3 ÷ 1/4 together. How many 1/4s fit in 3? Each whole holds 4 fourths.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- 3 ÷ 1/4 = 12 because it takes _____ jumps of _____ to get from 0 to _____.

Word bank Whole Number | Fraction | Reciprocal | Keep, Change, Flip | Quotient | 12 | fourth | jumps | 3 | four times | each whole

WATCH OUT!

A common mistake

A common mistake in Divide Whole Numbers by Fractions is doing 'Keep, Change' but skipping the 'Flip'

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Divide Whole Numbers by Fractions

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

How many jumps of 1/4 does it take to get from 0 to 2?



STEPS TO TRY
1. Each jump is exactly 1/4, and your goal is to land right on 2.
2. Count how many 1/4-jumps it takes to fill just ONE whole first.
3. It takes 4 jumps to reach 1. Keep jumping and counting from 1 up to 2.

SHOW YOUR THINKING

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 6 ÷ 1/3?

(A) 2

(B) 3

(C) 6/3

(D) 18

STEPS TO TRY

1. 6 ÷ 1/3 asks how many 1/3-size pieces fit into 6 wholes. Predict: bigger or smaller than 6?
2. Use the reciprocal shortcut: keep the 6, change ÷ to ×, and flip the divisor 1/3.
3. The flip turns 1/3 into 3/1. Set up 6 × 3 and finish the multiplication.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3 ÷ 1/5?

(A) 3/5

(B) 5/3

(C) 8

(D) 15

STEPS TO TRY

1. The total is 3 wholes and each piece is 1/5 — every whole holds several fifths.
2. Multiply by the reciprocal: keep 3, change ÷ to ×, and flip 1/5 upside down.
3. 1/5 flips to 5/1. Multiply 3 by 5 to count all the fifths.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 10 ÷ 1/4?

(A) 2.5

(B) 10/4

(C) 14

(D) 40

STEPS TO TRY

1. You're counting 1/4-size pieces inside 10 wholes — every whole holds more than one.
2. Keep 10, change ÷ to ×, and flip the divisor 1/4 before multiplying.
3. 1/4 becomes 4/1, so the problem turns into 10 × 4. Work that product out.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does 3 ÷ 1/4 = 12? How does the number line show this?

SENTENCE STARTER 3 ÷ 1/4 = 12 because it takes ___ jumps of ___ to get from 0 to ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help